

Charan Sai Soma

 LinkedIn  github.com/charansoma3001  <https://charansoma.vercel.app/>

 +44 7391 411283  charan.soma311@gmail.com  Edinburgh, UK, EH14 4AS

PERSONAL STATEMENT

AI Alignment & Robotics Engineer | Ex-Lead Developer (Managed 20+ Team)

MSc Artificial Intelligence candidate at Heriot-Watt University specializing in Embodied AI Safety and Edge Robotics. Bridges the gap between research and production engineering, combining 2+ years of leadership experience (architecting scalable systems for 300+ users) with advanced research in Red/Blue Teaming for robotic agents. Currently developing efficient Vision-Language Models (SmolVLM) for constrained edge hardware and validating safety protocols for human-robot collaboration using Google DeepMind's Concordia. Expert in bridging Sim2Real gaps using Unity, LeRobot, and local LLMs.

PROFESSIONAL EXPERIENCE

Graduate Research Engineer (Embodied AI & Safety)

September 2025 | Edinburgh, UK

The National Robotarium (Heriot-Watt University)

- **Adversarial Alignment (Red Teaming):** Engineering an "AI Control" framework using **Google DeepMind's Concordia**. Developed "Red Team" agents to generate contextual jailbreaks (e.g., convincing a robot to handle hazardous materials) and "Blue Team" safety filters to block them.
- **Edge AI Optimisation:** Ported multimodal models to Raspberry Pi 4 using custom **llama.cpp** forks. Optimising quantisation to reduce inference latency by **11%**, enabling real-time robotic decision-making without cloud dependency.
- **Robotic Control Policies:** Implemented end-to-end **Imitation Learning** pipelines using **Hugging Face LeRobot** and **ACT** (Action Chunking with Transformers) for the SO-101 robotic arm, bridging the Sim2Real gap.

Lead Software Developer

October 2024 – August 2025

XDLinux Space Labs

Hyderabad, India

- **Engineering Leadership:** Managed a team of **20+ developers** in building mission-critical satellite ground station software. Establishing CI/CD pipelines and code review standards that reduced critical production bugs by **33%**.
- **Local LLM Integration:** Fine-tuned open-source LLMs within a legacy C#/.NET environment to automate API documentation, reducing system integration time by **20%**.
- **Scalable Architecture:** Designed and deployed a secure Visitor Management System handling **300+ monthly users**, utilising Flutter, ASP.NET Core, and PostgreSQL with role-based access control (RBAC).

Web App Developer

June 2023 – September 2024

Tata Consultancy Services

Bengaluru, India

- **RAG Implementation & Agile Delivery:** Engineered a Retrieval-Augmented Generation (RAG) system using cloud-hosted LLMs to automate knowledge retrieval, collaborating within a high-performance agile team to deliver 100% of project milestones on time while significantly reducing query resolution time.
- **GenAI Tooling:** Developed a Generative AI assistant to streamline employee onboarding for **100+ new hires**, automating 10% of manual administrative workflows.

Software Developer (iOS & SAP Mobility)

January 2023 – June 2023

Incture (Internship)

Bengaluru, India

- **AR Navigation:** Built a Proof of Concept (PoC) for **Augmented Reality** fault location, helping technicians visualise maintenance targets and reducing timeouts by **45%**.

EDUCATION

Heriot-Watt University

September 2025 – September 2026

MSc in Artificial Intelligence (Distinction Expected)

Edinburgh, UK

Relevant Coursework:

- **Natural Language Processing:** Built **Bi-LSTM with Global Attention** from scratch (PyTorch) for NMT. Analyzed Word2Vec embedding scaling laws (Window Size vs. Corpus).
- **Data Mining (DMML):** Architected a **Hybrid Neural Recommendation System** (User Embeddings + CNN Content Encoder) solving the "Cold Start" problem.
- **Games Programming:** Implementing low-level *A Pathfinding** and **Physics Engines** in C#/Unity.

Extracurricular Engineering: Lead Developer for Unibots 2026 Autonomous Robot (Visual Servoing control loop & AprilTag localisation system (Sim2Real) to navigate dynamic arenas without GPS).

BNM Institute of Technology

August 2019 – May 2023

B.E. (Hons), Computer Science Engineering, CGPA: 8.92

Bengaluru, India

- **Result:** First Class with Distinction.
- **Capstone Project:** "Secure Gesture Control System with Face Anti-Spoofing."
 - Developed a dual-layer authentication system combining **CNN-based Face Liveness Detection** (to prevent photo spoofing) and **Hand Gesture Recognition**.
 - **Tech Stack:** OpenCV, TensorFlow Keras, Haar Cascades.
 - **Outcome:** Achieved **95% accuracy** in distinguishing real vs. fake faces and mapping custom gestures to automation tasks (e.g., smart lock control).

SKILLS

- **AI & Alignment:** AI Safety (Red/Blue Teaming), Imitation Learning (ACT/LeRobot), RLHF, Contextual Jailbreaking, Constitutional AI, RAG, LLM Fine-tuning (Unsloth).
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Computer Vision (OpenCV, CNNs), NLP (Transformers, LSTM, Word2Vec), Quantisation (llama.cpp).
- **Engineering & DevOps:** Python, C# (ASP.NET ⚡ Core), C++, Unity (Simulation), Webots (Simulation), Docker, CI/CD, Git, Raspberry Pi (Edge AI).
- **Frameworks:** Hugging Face, Google DeepMind Concordia, LangChain, Flutter, PostgreSQL.
- **Programming:** Python, JavaScript/TypeScript (Node.js, React, Next.js), SQL, Dart (Flutter), C++.
- **Services:** Firebase, Supabase, Microservices using Kafka, MongoDB and Zookeeper on Docker with RESTful APIs written in ASP.NET using C#.

AWARDS

Robotics Society Award, Heriot-Watt University

September 2025

Finished first in building the most efficient robotics solution within the shortest time.

Rapid Starter Award, Tata Consultancy Services

May 2024

Recognized for rapid learning curve and clear client/user communication skills.

Volunteering

Teach for India (Education Support), BNM Cycle Street (Sustainability), TCS Plantation Drives.